1. **Which two of the addresses below are available for host addresses on the subnet 192.168.15.19/28? (Select two answer choices)**
    A. 192.168.15.17
    B. 192.168.15.14
    C. 192.168.15.29
    D. 192.168.15.16
    E. 192.168.15.31
    F. None of the above

    Answer: A, C
    Explanation:
    - The network uses a 28bit subnet (255.255.255.240). This means that 4 bits are used for the networks and 4 bits for the hosts. This allows for 14 networks and 14 hosts (2n-2).
    - The last bit used to make 240 is the 4th bit (16) therefore the first network will be 192.168.15.16. The network will have 16 addresses (but remember that the first address Is the network address and the last address is the broadcast address).
    - In other words, the networks will be in increments of 16 beginning at 192.168.15.16/28. The IP address we are given is 192.168.15.19. Therefore the other host addresses must also be on this network.
    - Valid IP addresses for hosts on this network are: 192.168.15.17-192.168.15.30.

    Incorrect Answers:
    B. This is not a valid address for this particular 28 bit subnet mask. The first network address should be 192.168.15.16.
    D. This is the network address.
    E. This is the broadcast address for this particular subnet.

2. **How many subnetworks and hosts are available per subnet if you apply a /28 mask to the 210.10.2.0 class C network?**
    A. 30 networks and 6 hosts.
    B. 6 networks and 30 hosts.
    C. 8 networks and 32 hosts.
    D. 32 networks and 18 hosts.
    E. 16 networks and 14 hosts.
    F. None of the above

    Answer: E
    Explanation:
    - A 28 bit subnet mask (11111111.11111111.11111111.11110000) applied to a class C
    - network uses a 4 bits for networks, and leaves 4 bits for hosts.
    - Using the 2n-2 formula, we have 24-2 (or 2x2x2x2-2) which gives us 14 for the number of hosts, and the number of networks is 24 = 16.

    Incorrect Answers:
    G. This would be the result of a /29 (255.255.255.248) network.

H. This would be the result of a /27 (255.255.255.224) network.

I. This is not possible, as we must subtract two from the subnets and hosts for the network and broadcast addresses.

D. This is not a possible combination of networks and hosts.

3. **An organization was assigned the Class C network 199.166.131.0 from the ISP. If the administrator at TestKing were to subnet this class C network using the 255.255.255.224 subnet mask, how may hosts will they be able to support on each subnet?**

A.14
B.16
C.30
D.32
E.62
F.64

Answer: C

Explanation:

- The subnet mask 255.255.255.224 is a 27 bit mask (11111111.11111111.11111111.11100000).
- It uses 3 bits from the last octet for the network ID, leaving 5 bits for host addresses.
- We can calculate the number of hosts supported by this subnet by using the 2n-2 formula where n represents the number of host bits.
- In this case it will be 5. 25-2 gives us 30.

Incorrect Answers:

A. Subnet mask 255.255.255.240 will give us 14 host addresses.

B. Subnet mask 255.255.255.240 will give us a total of 16 addresses. However, we must still subtract two addresses (the network address and the broadcast address) to determine the maximum number of hosts the subnet will support.

D. Subnet mask 255.255.255.224 will give us a total of 32 addresses. However, we must still subtract two addresses (the network address and the broadcast address) to determine the maximum number of hosts the subnet will support.

E. Subnet mask 255.255.255.192 will give us 62 host addresses.

F. Subnet mask 255.255.255.192 will give us a total of 64 addresses. However, we must still subtract two addresses (the network address and the broadcast address) to determine the maximum number of hosts the subnet will support

4. **DAIICT consists of 5 departments A, B, C, D and E. Each department requires 7, 15, 13, 7 and 16 users respectively. You are a systems administrator and you've just acquired a new Class C address 199.166.131.0. Show the broadcast address, subnet address and usable host addresses for each of the departments.**

HINT: HERE THE REQUIREMENT FIRST IS TO FIND OUT SUFFICIENT BITS FOR SUBNETS

A. 255.255.255.128
B. 255.255.255.192
C. 255.255.255.224
D. 255.255.255.240
E. 255.255.255.248
F. 255.255.255.252

Answer: C

Explanation:

- The network currently consists of 5 subnets.
- We need to subnet the Class C network into at least 5 subnets.
- This requires that we use 3 bits for the network address.
- Using the formula 2n-2 we get 6.
- This also leaves us with 5 bits for hosts, which gives us 30 hosts.

Incorrect Answers:

G. Only 1 bit is required to give us 128 but 1 bit gives us 0 subnets. 2 bits are required to give us 192 but 2 bits gives us only 2 subnets. This is too few.

D. 4 bits are required to give us 240. This gives us 14 subnets. However we are left with 4 bits for hosts leaving us with 14 host addresses. Two of the networks require more than 14 hosts so this will not do.

E. 5 bits are required to give us 248. This gives us 30 subnets. However we are left with 3 bits for hosts leaving us with 6 host addresses. All the networks require more than 6 hosts so this will not do.

F. 6 bits are required to give us 252. This gives us 62 subnets. However we are left with 2 bits for hosts leaving us with 2 host addresses. This is too few